Statistics 280 — Practice Midterm 2B

Unofficial practice exam in the style of the W25 and W26 Midterm 2.
Covers Lectures 9–15 (Chapters 13–17): probability, Bayes' rule, counting, binomial and hypergeometric distributions, random variables, expected value, standard error, covariance, box models, and the normal approximation, plus R for probability and simulation.

Name _____ Start time / End time _____

- Answer all questions in the space provided and indicate where you have written additional work.
- Solutions without justification will be marked as wrong for all questions except for questions asking for R code. Show all work. A calculator command does not count as justification.
- Question 7 is an extra credit question (15 points).
- Express answers as fractions or as decimals rounded to 2 decimal places.
- The last page is a table of values from the normal distribution. Closed book, no notes, calculator allowed.
- Suggested time: 80 minutes. Total: 200 points + 15 extra credit.

Question	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**Total**
Points	30	30	40	30	40	30	15	200
Score								

1. (30 points) Each student in a class of 30 independently picks a 4-digit PIN at random. All 10,000 PINs from 0000 to 9999 are equally likely.

a) (4 points) What is the chance that two particular students pick the same PIN?

b) (7 points) What is the chance that all 30 PINs are different? Write it as a product, and also write a one-line R command that computes it.

c) (6 points) What is the chance that at least one of your 29 classmates picks the same PIN as you? Why is this much smaller than the chance that *some* two students share a PIN?

d) (7 points) Fill in the blanks so that the code finds the smallest class size n for which the chance that some two students share a PIN is at least 50%.

```
n = 1
while (prod((10000 - 0:(n - 1)) / 10000) > ______) {
  n = ______
}
print(______)
```

e) (6 points) The Chevalier de Méré problem (Chapter 14): which is more likely, at least one six in 4 rolls of a die, or at least one double-six in 24 rolls of a pair of dice? Find both chances.

2. (30 points) Roll two fair dice. Let A = "the first die is even", B = "the sum is 7", C = "the sum is 8", and D = "the second die shows 6".

a) (6 points) Find P(A), P(B), and P(A and B). Are A and B independent?

b) (6 points) Are A and C independent? Show your work.

c) (6 points) Are B and C mutually exclusive? Are they independent? Explain.

d) (5 points) Find P(A or D).

e) (7 points) Are B and D independent? Many people's intuition says no, since knowing the second die is a 6 seems to change the chance the sum is 7. Explain the result.

3. (40 points) Flip a fair coin. If it lands heads, roll one die. If it lands tails, roll two dice. Let T be the total number of spots rolled.

a) (6 points) What is P(T = 6)?

b) (6 points) Given that T = 6, what is the chance that the coin landed heads?

c) (4 points) Given that T ≥ 7, what is the chance that the coin landed heads?

d) (6 points) Given that T ≤ 3, what is the chance that the coin landed heads?

e) (6 points) Redo part b) using prior odds, data odds, and posterior odds.

f) (4 points) Give a frequency interpretation of your answer to part b).

g) (8 points) Write R code that simulates this experiment 100,000 times and estimates P(heads | T = 6). You may store the results of the runs in vectors.

4. (30 points) A club has 9 seniors and 6 juniors.

a) (4 points) How many different 5-person committees can be formed?

b) (6 points) If the committee is chosen at random, what is the chance that it has exactly 3 seniors?

c) (6 points) Instead, the club chooses a president, a vice president, and a treasurer (three different people), and then 2 general members from the rest. How many ways can this be done? If everything is chosen at random, what is the chance that the president and vice president are both seniors?

d) (6 points) All 15 members stand in a random line for a photo. Counting only the pattern of seniors and juniors, how many patterns are there? What is the chance that all 6 juniors stand next to each other?

e) (8 points) 7 of the 15 members have taken AP Statistics. A committee of 5 turns out to include 4 of them. If committees were chosen at random, what is the chance of 4 or more AP Statistics students on the committee? Give the formula with choose notation and an R command. Does the committee look unusual?

`R command: _____`

5. (40 points) A box contains four tickets: 1, 2, 2, 5. Draw two tickets without replacement. Let X be the first ticket and Y the second.

a) (6 points) Find P(X = 2 and Y = 2) and P(Y = 5).

b) (6 points) Find E(X) and SE(X). Are E(Y) and SE(Y) the same? Explain without listing the distribution of Y.

c) (8 points) Find E(XY) and Cov(X, Y). (Hint: XY is the product of two different tickets, and every ordered pair of different tickets is equally likely.)

d) (8 points) Find SE(X + Y) and SE(X − Y). Compare SE(X + Y) with its value if the draws were made with replacement, and explain the difference.

e) (8 points) Write R code that estimates Cov(X, Y) and SE(X + Y) from 100,000 simulated pairs of draws.

f) (4 points) Two random variables have covariance 0. Must they be independent? Give an example from lecture or the review questions.

6. (30 points) A roulette wheel has 38 pockets: 18 red, 18 black and 2 green. You bet $1 on red 100 times, winning $1 each time the ball lands on red and losing $1 otherwise.

a) (6 points) Describe the box model for your net winnings. Find the average and SD of the box.

b) (6 points) Find the expected value and standard error of your net winnings after 100 bets.

c) (8 points) Estimate the chance that you are ahead after 100 bets, using the normal approximation with the continuity correction on the number of bets you win. Also write the R command for the exact chance.

R command: _____

d) (5 points) Repeat the calculation of part c) for 10,000 bets.

e) (5 points) Use your answers to explain why the casino is happy to let players bet as many times as they like.

7. (15 points extra credit)

A box contains only two different numbers: a smaller number s on a fraction p_s of the tickets, and a bigger number b on a fraction $p_b = 1 - p_s$. Show that the SD of the box is $(b - s)\sqrt{p_b\, p_s}$.

A Normal Table

Height = height of the standard normal curve at z (percent per standard unit). Area = percentage of the area under the normal curve between −z and z.

z	Height	Area	z	Height	Area	z	Height	Area
0.00	39.89	0.00	**1.50**	12.95	86.64	**3.00**	0.443	99.730
0.05	39.84	3.99	**1.55**	12.00	87.89	**3.05**	0.381	99.771
0.10	39.70	7.97	**1.60**	11.09	89.04	**3.10**	0.327	99.806
0.15	39.45	11.92	**1.65**	10.23	90.11	**3.15**	0.279	99.837
0.20	39.10	15.85	**1.70**	9.40	91.09	**3.20**	0.238	99.863
0.25	38.67	19.74	**1.75**	8.63	91.99	**3.25**	0.203	99.885
0.30	38.14	23.58	**1.80**	7.90	92.81	**3.30**	0.172	99.903
0.35	37.52	27.37	**1.85**	7.21	93.57	**3.35**	0.146	99.919
0.40	36.83	31.08	**1.90**	6.56	94.26	**3.40**	0.123	99.933
0.45	36.05	34.73	**1.95**	5.96	94.88	**3.45**	0.104	99.944
0.50	35.21	38.29	**2.00**	5.40	95.45	**3.50**	0.087	99.953
0.55	34.29	41.77	**2.05**	4.88	95.96	**3.55**	0.073	99.961
0.60	33.32	45.15	**2.10**	4.40	96.43	**3.60**	0.061	99.968
0.65	32.30	48.43	**2.15**	3.96	96.84	**3.65**	0.051	99.974
0.70	31.23	51.61	**2.20**	3.55	97.22	**3.70**	0.042	99.978
0.75	30.11	54.67	**2.25**	3.17	97.56	**3.75**	0.035	99.982
0.80	28.97	57.63	**2.30**	2.83	97.86	**3.80**	0.029	99.986
0.85	27.80	60.47	**2.35**	2.52	98.12	**3.85**	0.024	99.988
0.90	26.61	63.19	**2.40**	2.24	98.36	**3.90**	0.020	99.990
0.95	25.41	65.79	**2.45**	1.98	98.57	**3.95**	0.016	99.992
1.00	24.20	68.27	**2.50**	1.75	98.76	**4.00**	0.013	99.9937
1.05	22.99	70.63	**2.55**	1.54	98.92	**4.05**	0.011	99.9949
1.10	21.79	72.87	**2.60**	1.36	99.07	**4.10**	0.009	99.9959
1.15	20.59	74.99	**2.65**	1.19	99.20	**4.15**	0.007	99.9967
1.20	19.42	76.99	**2.70**	1.04	99.31	**4.20**	0.006	99.9973
1.25	18.26	78.87	**2.75**	0.91	99.40	**4.25**	0.005	99.9979
1.30	17.14	80.64	**2.80**	0.79	99.49	**4.30**	0.004	99.9983
1.35	16.04	82.30	**2.85**	0.69	99.56	**4.35**	0.003	99.9986
1.40	14.97	83.85	**2.90**	0.60	99.63	**4.40**	0.002	99.9989
1.45	13.94	85.29	**2.95**	0.51	99.68	**4.45**	0.002	99.9991